Lifestyle Sustainability Assessment Using Machine Learning (EcoRate)

This presentation explores a novel approach to promoting sustainable lifestyles, using machine learning to provide personalized insights and recommendations.

Problem Statement

1 Importance of Sustainability

Climate change and environmental degradation are pressing issues, demanding a shift towards sustainable practices.

2 Challenges in Assessment

Individuals often lack clear understanding of their environmental impact, making it challenging to make informed decisions.

3 Improving Personal Impact

There's a need for a more accessible and effective method to assess and improve personal sustainability.

Objectives

Predictive Model

Develop a machine learning model that accurately predicts sustainability ratings based on individual lifestyle data.

Actionable Recommendations

Generate personalized recommendations tailored to users' individual needs, guiding them towards sustainable choices.

User-Friendly Dashboard

Visualize insights in a user-friendly and interactive dashboard, making it easy to understand and engage with sustainability data.

Methodology: Data Collection and Preprocessing

Data Sources

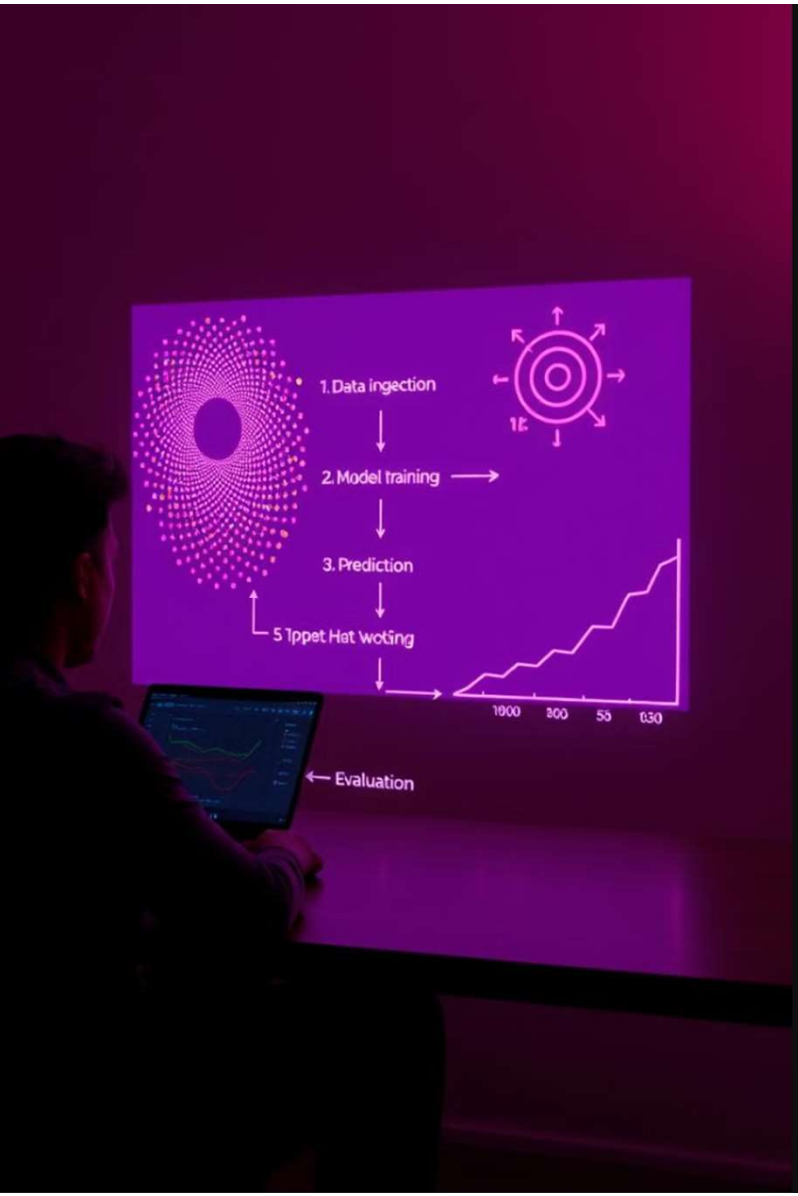
Data was collected from various sources, including user surveys, public databases, and wearable devices.

Data Cleaning

Data cleaning was performed to handle missing values, outliers, and inconsistencies, ensuring data quality.

Feature Engineering

Relevant features were extracted and transformed to enhance the model's performance, creating a comprehensive dataset.



Methodology: Machine Learning Model Development and Training

1 **Model Selection**

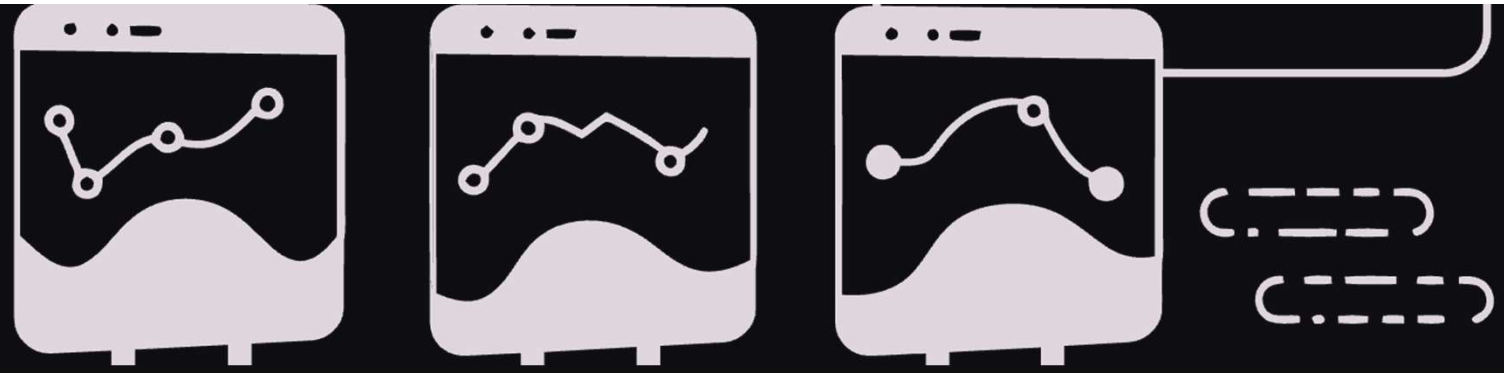
Several machine learning algorithms were explored, including regression models, classification models, and ensemble methods.

2 **Model Training**

The chosen model was trained on the processed dataset, optimizing its parameters to achieve high accuracy.

3 **Model Evaluation**

The trained model was evaluated using various metrics, including accuracy, precision, and recall, to assess its effectiveness.



Model Performance

Metric	Value
Mean Squared Error	0.73
R-squared	0.71

Test Case

Inputs: Urban, Plant-Based Diet, Renewable Energy, etc.

Predicted Sustainability Rating: 4.87

Discussion: Insights and Recommendations

1

Personalized Insights

The model provides personalized insights into users' sustainability scores, highlighting areas for improvement.

2

Actionable Recommendations

The system generates tailored recommendations, suggesting specific actions users can take to reduce their environmental impact.

3

Data-Driven Decision Making

Users can leverage these insights to make informed decisions and adopt more sustainable habits.

Conclusion

Sustainable Living

The model empowers individuals to live more sustainably by providing data-driven insights and guidance.

AI-Powered Solutions

Machine learning plays a crucial role in promoting sustainability by providing personalized and actionable solutions.

Collective Impact

By encouraging individual actions, this approach contributes to a broader shift towards a more sustainable future.

Future Work

1 Expand Data Sources

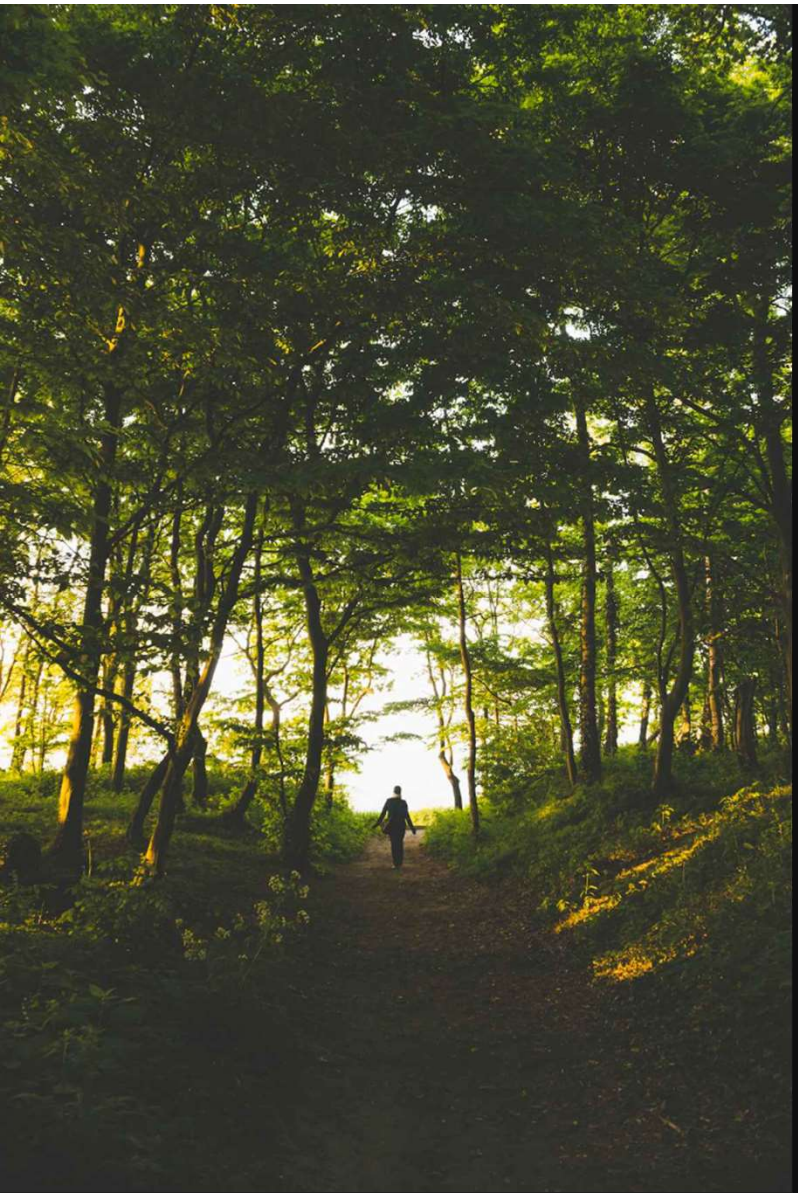
Include additional data sources to provide a more comprehensive view of user sustainability behavior.

2 Integrate Gamification

Implement gamification techniques to enhance user engagement and motivation for sustainable actions.

3 Develop Mobile App

Create a mobile application to provide a convenient and accessible platform for users to track their progress and receive recommendations.



Thank You!

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